Plan for Data Filtering:

1. Which teammates will specify, design, and implement this microservice

Nathan Zak and Mohana Vista will design this microservice; both they and Charles Gilbert will implement it.

2. Provide a link to the repository. Please make the link clickable.

https://github.com/m-vista/DU-DataFilter

3. Write at least three user stories for this Sprint. Provide your user stories and their functional and non-functional acceptance criteria (and associated quality attributes).

A. US-0X: Filter by Date Created

"As a user, I want to be able to filter my data by the date that I created it so that I can quickly see the newest and oldest entries."

Functional Requirements:
- Given I'm filtering data, when I filter for newest date created then I will see the same library of data as before but filtered in order from newest to oldest, and vice versa for filtering for oldest date created

B. US-0X: Filter by Integer

"As a user, I want to be able to filter my data by a specific integer so that I can quickly see any relevant entry."

Functional Requirements:
- Given I'm filtering data, when I enter a valid integer then I will only see items in my library of data whose attributes or contents contain that integer

C. US-0X: Filter by String

"As a user, I want to be able to filter my data by a specific string (keyword) so that I can quickly see any relevant entry."

Functional Requirements:
- Given I'm filtering data, when I enter a keyword(s) then I will only see

items in my library of data whose attributes or contents contain the keyword(s)

Non-functional Requirements:
- Provides rapid and precise search of large datasets at great ease

4. Take a screenshot that shows the user stories are in a Sprint Backlog in GitHub Projects.

(will take when we finalize user stories)

5. What kind of communication pipe will the microservice use? (e.g., text files, REST API)

ZeroMQ

6. How will other programs be able to request data from the microservice and what request parameters will be required? Give an example call using pseudocode or actual code.

Applications will request data using ZeroMQ REQ sockets and sending JSON strings as parameters

Example Pseudo:

```
create ZeroMQ request socket
    connect it to the filter service (port: xxxx)

build a request message with:
        (key) is (value)
        (key) is (value)
        ...

convert request message into JSON string
    send the JSON string through the socket, wait for a reply

    turn the reply from JSON back into data
    if the reply is "ok", use matching data
    else, show error message
```

7. How will other programs be able to receive data from the microservice and what data will the microservice provide? Give an example call using pseudocode or actual code.

A status field in a JSON string will either send “ok” and the correlating data or “error” and an explanatory message

Example Pseudo (turning JSON back into data):

...

```
if status is “ok”:
        read and display all relevant data
else, show error message
```